Applying Python Fundamentals and Libraries for Data Processing

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INTRODUCTION

This report outlines the process of loading, exploring, and processing a publicly available dataset as part of a data analysis task. The dataset was selected from an open-source data platform, Kaggle.
The dataset chosen is “Music & Mental Health Survey Results”
Link to the dataset is provided below:
https://www.kaggle.com/datasets/catherinerasgaitis/mxmh-survey-results

The main goal is to create a Python program that loads and processes the dataset using Pandas library. Tasks performed are:

1. Load the dataset
2. Display top 5 and bottom 5 rows
3. Display dataset size and data types of its columns
4. Filter out the data not needed
5. For loop to iterate through numeric columns
6. Calculating sum and average of numeric columns
7. For loop to iterate through string columns and convert them to lowercase
8. Counting group count
9. Generating summary statistics
10. Generating correlation table
11. Identifying duplicate rows and columns
12. Displaying count and percentage of null values

DATASET DESCRIPTION

The dataset selected is called "Music & Mental Health Survey Results" which has 33 columns and 736 rows. The columns (attributes) are as follows:

1. Timestamp (Date and time when form was submitted)
2. Age (Respondent's age)
3. Primary streaming service (Respondent's primary streaming service)
4. Hours per day (Number of hours the respondent listens to music per day)
5. While working (Does the respondent listen to music while studying/working?)
6. Instrumentalist (Does the respondent play an instrument?)
7. Composer (Does the respondent compose music?)
8. Fav genre (Respondent's favorite or top genre)
9. Exploratory (Does the respondent actively explore new artists/genres?)
10. Foreign languages (Does the respondent regularly listen to music with lyrics in a language they are not fluent in?)
11. BPM (Beats per minute of favorite genre)
12. Frequency [Classical] (How frequently does the respondent listen to classical music?)
13. Frequency [Country] (How frequently does the respondent listen to country music?)
14. Frequency [EDM] (How frequently does the respondent listen to EDM music?)
15. Frequency [Folk] (How frequently does the respondent listen to folk music?)
16. Frequency [Gospel] (How frequently does the respondent listen to gospel music?)
17. Frequency [Hip hop] (How frequently does the respondent listen to hip hop music?)
18. Frequency [Jazz] (How frequently does the respondent listen to jazz music?)
19. Frequency [K pop] (How frequently does the respondent listen to kpop music?)
20. Frequency [Latin] (How frequently does the respondent listen to latin music?)
21. Frequency [Lofi] (How frequently does the respondent listen to lofi music?)
22. Frequency [Metal] (How frequently does the respondent listen to metal music?)
23. Frequency [Pop] (How frequently does the respondent listen to pop music?)
24. Frequency [R&B] (How frequently does the respondent listen to R&B music?)
25. Frequency [Rap] (How frequently does the respondent listen to rap music?)
26. Frequency [Rock] (How frequently does the respondent listen to rock music?)
27. Frequency [Video game music] (How frequently does the respondent listen to video game music?)
28. Anxiety (Self-reported anxiety, 0-10)

29. Depression (Self-reported depression, 0-10)
30. Insomnia (Self-reported insomnia, 0-10)
31. OCD (Self-reported OCD, 0-10)
32. Music effects (Does music improve/worsen respondent's mental health conditions?)
33. Permissions (Permission to publicize data)

A brief overlook of the dataset:

Timestam	Age	Primary st	Hours per	While wor	Instrumer	Compose	Fav genre	Explorator	Foreign lar	BPM	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Frequency	Anxiety	Depressio	Insomnia	OCD	Music effects	Permissions
8/27/202	18	Spotify	3	Yes	Yes	Yes	Latin	Yes	Yes	156	Rarely	Never	Rarely	Never	Never	Sometime	Never	Very frequ	Very frequ	Rarely	Never	Very frequ	Sometime	Very frequ	Never	Sometime	3	0	1	0		I understand.
8/27/202	63	Pandora	1.5	Yes	No	No	Rock	Yes	No	119	Sometime	Never	Never	Rarely	Sometime	Rarely	Very frequ	Rarely	Sometime	Rarely	Never	Sometime	Sometime	Rarely	Very frequ	Rarely	7	2	2	1		I understand.
8/27/202	18	Spotify	4	No	No	No	Video gam	No	Yes	132	Never	Never	Very frequ	Never	Never	Rarely	Rarely	Very frequ	Never	Sometime	Sometime	Rarely	Never	Rarely	Rarely	Very frequ	7	7	10	2	No effect	I understand.
8/27/202	61	YouTube M	2.5	Yes	No	Yes	Jazz	Yes	Yes	84	Sometime	Never	Never	Rarely	Sometime	Never	Very frequ	Sometime	Very frequ	Sometime	Never	Sometime	Sometime	Never	Never	Never	9	7	3	3	Improve	I understand.
8/27/202	18	Spotify	4	Yes	No	No	R&B	Yes	No	107	Never	Never	Rarely	Never	Rarely	Very frequ	Never	Very frequ	Sometime	Sometime	Never	Sometime	Very frequ	Very frequ	Never	Rarely	7	2	5	9	Improve	I understand.
8/27/202	18	Spotify	5	Yes	Yes	Yes	Jazz	Yes	Yes	86	Rarely	Sometime	Never	Never	Never	Sometime	Very frequ	Very frequ	Rarely	Very frequ	Rarely	Very frequ	Very frequ	Very frequ	Very frequ	Never	8	8	7	7	Improve	I understand.
8/27/202	18	YouTube M	3	Yes	Yes	No	Video gam	Yes	Yes	66	Sometime	Never	Rarely	Sometime	Rarely	Rarely	Sometime	Never	Rarely	Rarely	Rarely	Rarely	Rarely	Never	Never	Sometime	4	8	6	0	Improve	I understand.
8/27/202	21	Spotify	1	Yes	No	No	K pop	Yes	Yes	95	Never	Never	Rarely	Never	Never	Very frequ	Rarely	Very frequ	Never	Sometime	Never	Sometime	Sometime	Rarely	Never	Rarely	5	3	5	3	Improve	I understand.
8/27/202	19	Spotify	6	Yes	No	No	Rock	No	No	94	Never	Very frequ	Never	Sometime	Never	Never	Never	Never	Never	Never	Very frequ	Never	Never	Never	Very frequ	Never	2	0	0	0	Improve	I understand.
8/27/202	18	I do not us	1	Yes	No	No	R&B	Yes	Yes	155	Rarely	Rarely	Rarely	Rarely	Sometime	Rarely	Rarely	Never	Rarely	Rarely	Never	Sometime	Sometime	Rarely	Sometime	Sometime	2	2	5	1	Improve	I understand.
8/27/202	18	Spotify	3	Yes	Yes	No	Country	Yes	No		Never	Very frequ	Never	Never	Never	Never	Never	Never	Never	Never	Never	Rarely	Rarely	Never	Rarely	Never	7	7	4	7	No effect	I understand.
8/27/202	19	YouTube M	8	Yes	No	No	EDM	Yes	No	125	Rarely	Never	Very frequ	Never	Never	Sometime	Rarely	Rarely	Rarely	Rarely	Never	Rarely	Rarely	Sometime	Rarely	Rarely	1	0	0	1	Improve	I understand.
8/27/2022	23:04:00	Spotify	3	Yes	No	No	Hip hop	Yes	Yes		Rarely	Never	Rarely	Never	Never	Very frequ	Never	Sometime	Never	Very frequ	Never	Sometime	Sometime	Rarely	Rarely	Never	9	3	2	7	Improve	I understand.
8/27/202	19	Spotify	2	Yes	No	No	Country	Yes	No	88	Never	Very frequ	Rarely	Sometime	Never	Sometime	Never	Never	Never	Rarely	Never	Rarely	Never	Very frequ	Never	Never	2	1	2	0	Improve	I understand.
8/27/202	18	Spotify	4	Yes	Yes	No	Jazz	Yes	Yes	148	Very frequ	Rarely	Never	Never	Never	Never	Very frequ	Never	Sometime	Rarely	Sometime	Sometime	Never	Never	Sometime	Rarely	6	4	7	0	Improve	I understand.
8/27/202	17	Spotify	2	No	No	No	Pop	Yes	Yes		Rarely	Rarely	Never	Never	Never	Sometime	Rarely	Rarely	Rarely	Rarely	Rarely	Very frequ	Rarely	Sometime	Sometime	Rarely	7	5	4	1	Worsen	I understand.
8/27/202	16	Spotify	8	Yes	No	No	Hip hop	Yes	Yes	103	Never	Never	Never	Never	Never	Very frequ	Never	Never	Never	Never	Never	Never	Sometime	Very frequ	Never	Rarely	8	8	4	3	Improve	I understand.
8/27/202	16	Spotify	12	Yes	No	Yes	Hip hop	Yes	Yes	120	Rarely	Never	Sometime	Rarely	Never	Very frequ	Rarely	Never	Never	Never	Sometime	Sometime	Rarely	Sometime	Very frequ	Never	5	7	10	0	Improve	I understand.
8/27/202	17	Spotify	24	Yes	No	No	Rap	Yes	No	99	Rarely	Never	Never	Never	Never	Sometime	Sometime	Sometime	Rarely	Never	Sometime	Rarely	Sometime	Very frequ	Very frequ	Never	7	5	0	3	Improve	I understand.

Figure 1: Dataset overlook

DATASET EXPLORATION AND PROCESSING

1. Dataset size and data types of its columns

```
number of rows and columns in the dataset (size of the dataset):  (736, 33)

columns and their data types:
 Timestamp                        object
Age                              float64
Primary streaming service         object
Hours per day                    float64
While working                     object
Instrumentalist                   object
Composer                          object
Fav genre                         object
Exploratory                       object
Foreign languages                 object
BPM                              float64
Frequency [Classical]             object
Frequency [Country]               object
Frequency [EDM]                   object
Frequency [Folk]                  object
Frequency [Gospel]                object
Frequency [Hip hop]               object
Frequency [Jazz]                  object
Frequency [K pop]                 object
Frequency [Latin]                 object
Frequency [Lofi]                  object
Frequency [Metal]                 object
Frequency [Pop]                   object
Frequency [R&B]                   object
Frequency [Rap]                   object
Frequency [Rock]                  object
Frequency [Video game music]      object
Anxiety                          float64
Depression                       float64
Insomnia                         float64
OCD                              float64
Music effects                     object
Permissions                       object
```

Figure 2: Dataset size and column data types

Dataset has 33 columns and 736 rows.

Data types:

object – categorical value stored as text

float64 – floating point number data type

2. Filtered out data which is not required

```
required dataset columns and column types:
 Age                              float64
Primary streaming service         object
Hours per day                    float64
Frequency [Classical]             object
Frequency [Country]               object
Frequency [EDM]                   object
Frequency [Folk]                  object
Frequency [Gospel]                object
Frequency [Hip hop]               object
Frequency [Jazz]                  object
Frequency [K pop]                 object
Frequency [Latin]                 object
Frequency [Lofi]                  object
Frequency [Metal]                 object
Frequency [Pop]                   object
Frequency [R&B]                   object
Frequency [Rap]                   object
Frequency [Rock]                  object
Frequency [Video game music]      object
Anxiety                          float64
Depression                       float64
Insomnia                         float64
OCD                              float64
Music effects                     object
dtype: object
size of the required dataset:  (736, 24)
```

Figure 3: Required dataset

Required dataset has 24 columns and 736 rows.

All further operations will be performed on this required dataset.

3. Summary statistics

```
              Age  Hours per day     Anxiety  Depression    Insomnia  \
count  735.000000     736.000000  736.000000  736.000000  736.000000
mean    25.206803       3.572758    5.837636    4.796196    3.738451
std     12.054970       3.028199    2.793054    3.028870    3.088689
min     10.000000       0.000000    0.000000    0.000000    0.000000
25%     18.000000       2.000000    4.000000    2.000000    1.000000
50%     21.000000       3.000000    6.000000    5.000000    3.000000
75%     28.000000       5.000000    8.000000    7.000000    6.000000
max     89.000000      24.000000   10.000000   10.000000   10.000000

              OCD
count  736.000000
mean     2.637228
std      2.842017
min      0.000000
25%      0.000000
50%      2.000000
75%      5.000000
max     10.000000
```

Figure 4: Summary statistics

Age:

- Mean age (25.20) suggests a young population, with a wide age range (10 to 89 years)
- The median age (21 years) being lower than the mean indicated a right-skewed distribution.

Listen Time (Hours per day):

- Mean listen time (3.57 hours/day) indicates moderate daily listen time
- Median of 3 hours shows that half of the participants spend 3 hours or less listening to music, but some report very high listen time (up to 24 hours)

Anxiety:

- Mean (5.84) indicates moderate anxiety
- Median score (6) shows that most respondents have moderate to high anxiety levels

Depression:

- Mean score (4.8) suggests moderate depression levels
- Median score (5) suggests a relatively symmetric distribution of depression levels

Insomnia:

- Mean score (3.74) indicates moderate insomnia
- Median score (3) shows that half of the participants have mild insomnia symptoms

OCD:

- Mean score (2.64) suggests lower OCD symptoms compared to other mental health issues
- Median score (2) indicates that many participants report little to no OCD symptoms

4. Correlation table analysis

	Hours per day	Anxiety	Depression	Insomnia	OCD
Hours per day	1.000000	0.049319	0.110527	0.141821	0.118729
Anxiety	0.049319	1.000000	0.519969	0.292669	0.348350
Depression	0.110527	0.519969	1.000000	0.378996	0.196988
Insomnia	0.141821	0.292669	0.378996	1.000000	0.226354
OCD	0.118729	0.348350	0.196988	0.226354	1.000000

Figure 5: Correlation table

- The correlation table shows weak positive relationship between listen time and the mental health conditions, suggesting that listening to music has minimal impact on these issues.
- The strongest relationships are found between the mental health conditions themselves, particularly between Anxiety and Depression (0.52) and between Depression and Insomnia (0.38), indicating that these conditions tend to co-occur or are closely linked.

DATA QUALITY

```
null values for each column:
 Age                                 1
Primary streaming service           1
Hours per day                       0
Frequency [Classical]               0
Frequency [Country]                 0
Frequency [EDM]                     0
Frequency [Folk]                    0
Frequency [Gospel]                  0
Frequency [Hip hop]                 0
Frequency [Jazz]                    0
Frequency [K pop]                   0
Frequency [Latin]                   0
Frequency [Lofi]                    0
Frequency [Metal]                   0
Frequency [Pop]                     0
Frequency [R&B]                     0
Frequency [Rap]                     0
Frequency [Rock]                    0
Frequency [Video game music]        0
Anxiety                             0
Depression                          0
Insomnia                            0
OCD                                 0
Music effects                       8
dtype: int64

total null values: 10

percentage of null values:  1.358695652173913
```

```
number of duplicate row entries:  0
number of duplicate column entries:  0
```

Figure 6: Null value and duplicate value count

- Total number of null values = 0, i.e., 1.36% of the whole dataset, which is low.
- There are no duplicate values in either rows or columns.
- Therefore, the dataset is of good quality and is reliable.

CONCLUSION

In this data analysis task, we successfully explored and processed the “Music & Mental Health Survey Results” dataset, consisting of 33 columns and 736 rows. The dataset showed moderate levels of anxiety, depression, insomnia, and OCD among respondents, with young population predominantly listening to music for 3-4 hours daily.

The correlation analysis revealed weak relationships between music listening time and mental health conditions, but stronger connections between conditions themselves, particularly anxiety and depression.

The dataset was found to be of good quality, with minimal missing values (1.36%) and no duplicates, ensuring its reliability for analysis.

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